

Doubly Periodic Functions and Elliptic Curves

Toyesh K. Jayaswal

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1 Introduction

An very interesting object of study in modern mathematics is the elliptic curve. An elliptic curve can be described as the set of (projective) solutions to the polynomial $y^2 = x^3 + ax + b$ where the polynomial on the right side doesn't have any repeated roots. There are many equivalent descriptions of an elliptic curve, and one of these is that an elliptic curve is a smooth curve of genus one. The genus is a topological concept, and it isn't initially clear how to relate this polynomial equation to topology. The answer turns out to be that if you plot out all complex solutions, the curve traced out will look like a torus. So it's easy to see how the genus comes into play: all we need to do is imagine the solution set in $\mathbb{C}^2$ and see very clearly that it looks like a donut! Just one small issue: $\mathbb{C}^2$ is two copies of $\mathbb{C}$, so in total it's four dimensional and we need to visualize four dimensional space. Hence, we cannot graph the solutions and easily see the topology. In this talk, we will give a convincing argument as to why the solutions to an elliptic curve should look like a torus. We start with something that seems completely unrelated. A word of warning: there are inevitably some things that we will have to blackbox, mainly some theorems in complex analysis. Luckily, theorems in complex analysis are really simple to state, but surprisingly powerful. For example, one of the main theorems we will use is that if a holomorphic (i.e. complex differentiable, pronounced "nice") function $f : \mathbb{C} \rightarrow \mathbb{C}$ is bounded i.e. there is some M so that $|f(z)| < M$ for all $z \in \mathbb{C}$, then $f(z)$ must be a constant function! Everything we assume going forward will be meromorphic (holomorphic plus possibly poles, pronounced "nice"). An analogy of this statement is that if a polynomial has no zeros, it must be constant.

2 Periodic Functions

$\sin(x)$ has the nice property that it is periodic since $\sin(x + 2\pi) = \sin(x)$. We can extend $\sin$ to the complex numbers by using Euler's formula, however, we don't have any periodicity in the imaginary direction. An interesting problem to look at is that $\mathbb{C}$ has 2 real directions, so can we find some function that is doubly periodic? In other words, can we find two complex numbers ω_1, ω_2 (such that their ratio isn't real so that they are linearly independent over $\mathbb{R}$) and some function $\wp(z)$ so that $\wp(z + \omega_1) = \wp(z + \omega_2) = \wp(z)$ for all $z \in \mathbb{C}$. An easy function that satisfies this property is the constant function $\wp(z) = C$, but we want to see if we can find a more nontrivial function. Let's study some properties of such a function, were it to exist. We can consider the lattice $L = \{m\omega_1 + n\omega_2 | m, n \in \mathbb{Z}\}$ and we have that $\wp(z + \omega) = \wp(z)$ for all $\omega \in L$ (exercise). Hence, we can see that the function on $\mathbb{C}$ is determined by the values the function takes on a fundamental parallelogram of our lattice. This is because we can tile the plane with copies of the fundamental parallelogram and these translations do not change the value of our function. Here, we obtain our first important observation: $\wp(z)$ cannot be bounded on the fundamental parallelogram (including the boundary). This is because if it's bounded on the fundamental parallelogram, we can use our translation trick to show that the function must be bounded on the entire complex plane, and hence from the theorem stated in the introduction, it must be constant!

This means that there must be some point in the fundamental parallelogram where the function "blows up to infinity" like z^{-1} does at 0. If $f(z)$ has such behavior so that $f(z)$ looks like $(z - p)^{-n}$ for z near p , we say f has a pole of order n at p . Hence, we conclude that there must be some point inside the fundamental parallelogram where $\wp$ has a pole. If you feel a bit skeptical that we must have a point where $\wp$ blows up just because it isn't bounded, the argument holds because the fundamental parallelogram with boundary is a compact set. In fact, it can be shown that there must be at least two poles within the fundamental parallelogram. This unfortunately needs to be one of the things that we blackbox, but if you know complex analysis, consider it an exercise to show that the sum of the residues of $\wp$ within the fundamental parallelogram (or some translate of it so that the poles aren't on the boundary) is 0.

We now have a function that must have at least two poles within the fundamental parallelogram. Let's try to use this to take a guess at a function which would be doubly periodic. Let's put both poles at 0 so that near 0, $\wp(z) = z^{-2} + a_{-1}z^{-1} + a_0 + a_1z + \dots$. This is part of the blackboxed exercise above, but we must also have that the coefficient of z^{-1} is 0 so that our function looks like $\wp(z) = z^{-2} + a_0 + a_1z + \dots$. Let's try to force our function to be doubly periodic as follows: define

$$\wp(z) = \sum_{\omega \in L} \frac{1}{(z - \omega)^2}.$$

Why choose this? The behavior at 0 tells us that near $z = \omega \in L$, $\wp$ must look like $(z - \omega)^{-2}$. If we were to look at $\wp(z + \omega')$ for $\omega' \in L$, we have that $\wp(z + \omega') = \sum_{\omega \in L} \frac{1}{(z + \omega' - \omega)^2}$. However, by rearranging our sum, we have that $\{\omega - \omega' | \omega \in L\} = L$ (translating by ω' forms a bijection of elements of the lattice). This means that $\wp(z + \omega') = \sum_{\omega \in L} \frac{1}{(z - (\omega - \omega'))^2} = \sum_{\omega \in L} \frac{1}{(z - \omega)^2} = \wp(z)$. We've done it! We constructed a function that is doubly periodic!

Except there's a small problem: our best friend convergence issues comes to haunt us. The sum converges away from the poles, but is "conditionally convergent" instead of "absolutely convergent" which means that the order in which we take our sum can actually change the answer. A series $f(z) = \sum a_k z^k$ is said to be absolutely convergent at z if $\sum |a_n z^n|$ converges. It can be shown that if a sum with infinitely many terms is absolutely convergent, its value does not change by rearranging the summands. However, this is not true in general (check out Riemann series theorem if you're interested). Let's give a heuristic argument as to why the sum does not converge uniformly. If $z \approx 0$, we have that $\wp(z) \approx \sum_{\omega \in L} \frac{1}{\omega^2}$. Obviously the $\omega = 0$ term will cause divergence, so let's check if the remaining terms converge or not. For uniform convergence, we must then check if $\sum_{\omega \in L - 0} \frac{1}{|\omega|^2}$ converges. Let $L_k = \{a\omega_1 + b\omega_2 | a, b \in \mathbb{Z}, |a|, |b| \leq k\}$ (if necessary, change ω_1, ω_2 to be the generators of the fundamental parallelogram). For example L_0 is just the origin and L_1 has 9 points. We have that $|L_k - L_{k-1}| = 8k$ (exercise). Hence, we can write our sum as $\lim_{N \rightarrow \infty} \sum_{\omega \in L_N - 0} \frac{1}{|\omega|^2} = \lim_{N \rightarrow \infty} \sum_{k=1}^N \sum_{\omega \in L_k - L_{k-1}} \frac{1}{|\omega|^2}$. We see that at step L_k , we add new terms on the order of k while the size of our terms also increase on the order of k (why?). Hence, our sum is heuristically

$$\lim_{N \rightarrow \infty} \sum_{k=1}^N \sum_{\omega \in L_k - L_{k-1}} \frac{1}{|\omega|^2} \sim \lim_{N \rightarrow \infty} \sum_{k=1}^N \frac{k}{k^2} = \sum_{k=1}^{\infty} \frac{1}{k},$$

which is the famous diverging harmonic series.

To fix this, it turns out that all we have to do is subtract off these problematic terms ω^{-2} . The function

$$\wp(z) = \frac{1}{z^2} + \sum_{\omega \in L - 0} \left(\frac{1}{(z - \omega)^2} - \frac{1}{\omega^2} \right)$$

converges uniformly so that we can rearrange terms, and is even doubly periodic! If you don't believe that it is doubly periodic, consider the function $\sum_{\omega \in L} (z - \omega)^{-3}$. This function, by the same argument as earlier, is doubly periodic, and can be shown to be uniformly convergent so that it is actually valid

to rearrange the terms. The function $\wp(z)$ is obtained by integrating this function from 0 to z (the term z^{-3} is integrated separately, hence the absence from the sum, you can fix this by choosing the initial integration point to not be in L , but that would just change $\wp(z)$ by a constant). This function is very special and is known as the Weierstrass $\wp$ -function (pronounced Weierstrass p function).

Let's do some manipulations on $\wp(z)$. First, let's try and compute out its Laurent series. A Laurent series is essentially a power series with maybe some negative components. For example, $\frac{1}{z(1-z)} = \frac{1}{z} + \frac{1}{1-z} = \frac{1}{z} + 1 + z + z^2 + \dots$

Exercise (Power Series Expansion)

Show that

$$\wp(z) = \frac{1}{z^2} + \sum_{k=1}^{\infty} (2k+1)G_{2k+2}z^{2k}$$

with

$$G_k = \sum_{\omega \in L-0} \frac{1}{\omega^n} = \sum_{a,b \in \mathbb{Z}, ab \neq 0} \frac{1}{(a\omega_1 + b\omega_2)^n}.$$

Do this by expanding $(z-\omega)^{-2}$ into a power series for all ω and combine terms which share a common power of z . As an exercise, why is there no constant coefficient and no odd coefficient. Note that $\wp(z)$ is an even function since its series expansion only has even exponents (this could also have been seen from the original expression by reordering terms). These functions G_k are examples of what are called modular forms, which are functions with some very beautiful properties and are very important in modern research, but this is beyond these notes.

3 Relation to Elliptic Curves

We now look at some things relating to $\wp(z)$ and eventually get to showing that an elliptic curve is homeomorphic topologically to (pronounced "looks like") a torus. Let's analyze the derivative of the $\wp$ -function, $\wp'(z)$. Differentiating term by term have that

$$\wp'(z) = -\frac{2}{z^3} + \sum_{k=1}^{\infty} (2k+1)(2k)G_{2k+2}z^{2k-1} = -\frac{2}{z^3} + 6G_4z + 20G_6z^3 + \dots$$

Note that $\wp'(z)$ is an odd function. We can construct an even function by squaring it:

$$[\wp'(z)]^2 = \frac{4}{z^6} - \frac{24G_4}{z^2} - 80G_6 + \dots$$

We can do something interesting and try to cancel the poles from this function using $\wp(z)$. First, $[\wp(z)]^3$ has a pole of order 6. $\wp(z)$ starts as $\wp(z) = \frac{1}{z^2} + 3G_4z^2 + 5G_6z^4 + \dots$ so we can cancel out the first term by:

$$[\wp'(z)]^2 - 4[\wp(z)]^3 = \left(\frac{4}{z^6} - \frac{24G_4}{z^2} - 80G_6 + \dots \right) - 4 \left(\frac{1}{z^6} + \frac{9G_4}{z^2} + 15G_6 + \dots \right) = -\frac{60G_4}{z^2} - 140G_6 + \dots$$

Now, we can cancel the second order pole using $\wp(z)$ and we have

$$[\wp'(z)]^2 - 4[\wp(z)]^3 + 60G_4\wp(z) = -\frac{60G_4}{z^2} - 140G_6 + \dots + 60G_4 \left(\frac{1}{z^2} + 3G_4z^2 + \dots \right) = -140G_6 + \dots$$

Hence, we have the power series expansion $[\wp'(z)]^2 - 4[\wp(z)]^3 + 60G_4\wp(z) = -140G_6 + \dots$ Now, here is where the magic happens: on the left hand side, we have taken derivatives, powers, and sums of the doubly periodic function $\wp(z)$. The only defining property of a doubly periodic function is that if

you shift the input by either of the periods, the value of the function will be left invariant. Verify that this property still holds if you take the derivative of such a function, the power of such a function, or the sum of two such functions. Hence, we have that $[\wp'(z)]^2 - 4[\wp(z)]^3 + 60G_4\wp(z)$ is a doubly periodic function. But since there's no nonzero negative coefficients of z , we have that it is a doubly periodic function with no poles in the fundamental parallelogram. But we showed such a function must be constant! Hence, a priori, we didn't know what the coefficients of $z, z^2, \dots$ were for our equation above, but this tells us they must all be 0, and we have the identity $[\wp'(z)]^2 - 4[\wp(z)]^3 + 60G_4\wp(z) = -140G_6$, or in other words $[\wp'(z)]^2 = 4[\wp(z)]^3 - 60G_4\wp(z) - 140G_6$. If you don't see what's so special about this equation, we have that $z \rightarrow (\wp(z), \wp'(z))$ is a parametrization for the elliptic curve $y^2 = 4x^3 - 60G_4x - 140G_6$, and with a change of variables, the coefficient of x can be made to 1.

It can be shown that the roots of the polynomial on the right has no repeated roots (the discriminant can be shown to always be nonzero by studying G_4, G_6). Moreover, we can show that for any a, b , we can find a lattice in $\mathbb{C}$ such that $G_4 = a$ and $G_6 = b$. These involve some more technical calculations which we do not go into here, but they are described very nicely in Apostol's book. This means that given any elliptic curve, we can parametrize it using $\wp$ and $\wp'$! This is analogous to how quadratic equations can be parametrized by suitable multiples of $\sin$ and its derivative, or $\sinh$ and its derivative (which are really the same over complex numbers). In all of these examples, we have some analytic object which explicitly parametrizes our elliptic curve.

Now, to show that the parametrization actually covers every point on the elliptic curve, we must at least show that $\wp(z)$ takes every value in $\mathbb{C}$ at least once inside the fundamental parallelogram, since for every $x \in \mathbb{C}$, there is a solution for y of our elliptic curve. We do this with another black boxed result using complex analysis: the number of poles and the number of zeros of a doubly periodic function must be the same within a fundamental parallelogram (shifted if necessary to avoid poles and zeros on the boundary). If you know complex analysis, this is also a worthwhile exercise. We have that in the fundamental parallelogram, we have a pole of order 2 at each corner, but we can move our parallelogram by a little bit to get a parallelogram that only contains a pole of order 2 at the origin. Note that because of periodicity, studying the values of $\wp$ on this shifted parallelogram is equivalent to studying the values on the original (why?). Since we have two poles inside this fundamental parallelogram, we conclude that we must also have exactly two zeros inside the parallelogram. But, $\wp(z) - c$ for any constant c must also have two zeros! This is because adding a constant to a doubly periodic function keeps it doubly periodic, and doesn't change the number of poles. $\wp(z) - c = 0$ is equivalent to $\wp(z) = c$, so we have that $\wp(z)$ takes every value on $\mathbb{C}$ at least once.

Now, for every value of $x \in \mathbb{C}$, y has two values unless x is a root of the polynomial on the right, in which case y just has the one solution 0. Since $\wp(z)$ has two zeros in the fundamental parallelogram, we have two cases: the two zeros take places in different places, or there is a zero is a double root of $\wp(z)$. Let's analyze the second case first: since it's a double root, we have that $\wp'(z) = 0$. Since $\wp'(z)$ has a pole of order 3, it must have 3 zeros. We can show these to be at $\omega_1/2, \omega_2/2, (\omega_1 + \omega_2)/2$ where ω_1, ω_2 are the basis vectors for the fundamental parallelogram (exercise; use the fact that $\wp'$ is an odd function). Now, let's say $\wp'(z) \neq 0$. We have that $\wp(z) = c = \wp(-z)$, but and both $y = \wp'(z)$ and $y = \wp'(-z)$ satisfy $y^2 = 4c^3 - 60G_4c - 140G_6$ because of the parametrization. If these two values were equal, it would be a double root. Hence, we have the same behavior as the elliptic curve where y takes two values for every value of x except when x is a root of the polynomial, in which case y takes the value 0 with multiplicity 2. Moreover, we can actually compute the roots of the polynomial $4c^3 - 60G_4c - 140G_6$ to be $x = \wp(\omega_1/2), \wp(\omega_2/2), \wp((\omega_1 + \omega_2)/2)$, because these inputs are precisely the zeros of $\wp'(z)$.

In fact, we not only showed that the Weierstrass $\wp$ -function forms a surjective parametrization of the elliptic curve, but also that the elliptic curve is bijective with our fundamental parallelogram! Check to make sure this makes sense: to every x value, there's two z 's from the fundamental parallelogram for which $\wp(z) = x$ except at the roots where there's only one, and this is exactly the same as the count of the points on the elliptic curve for every x value. Moreover, the parametrization is continuous so that nearby values of z map to nearby values of the solutions of the elliptic curve. Hence, the two objects (fundamental parallelogram and the solutions of the elliptic curve) really will look

the same. Here's the beautiful thing: the fundamental parallelogram has some more structure. The opposite sides should actually be identified because the $\wp$ -function is periodic on the lattice and the points on either edge are translates of one another. But a parallelogram with opposite edges identified is topologically the same as a square with opposite edges identified, but this is the famous description of a torus! Hence, we have shown that the solutions to an elliptic curve in $\mathbb{C}^2$ is in fact a torus! There is one small technical point which is you must projectivize to account for the point at infinity to account for the point $z = 0$ as an input for the $\wp$ -function. This entire construction is analogous to how we can use the parametrization for the $x^2 + y^2 = 1$ with $\sin(t), \cos(t)$ to show that the real solutions to $x^2 + y^2 = 1$ is homeomorphic to $\mathbb{R}/\mathbb{Z}$ since $\sin$ is periodic, and this proves it's a circle. This is a fact that we are quite comfortable with, but technically it's very much not trivial, and this outlines another example of how some analytic functions can be used to describe the topological behavior of an algebraic equation.

The theory is very deep here. This description essentially says that elliptic curves over $\mathbb{C}$ are easy to solve since we just have to find the corresponding lattice and compute the $\wp$ function. However, you may have heard that solving an elliptic curve over $\mathbb{Q}$ or other fields is hard. This makes sense since finding a point over $\mathbb{Q}$ corresponds to finding a value of z for which both $\wp(z)$ and $\wp'(z)$ are rational, similar to how rational points on a circle correspond to angles for which $\sin(\theta)$ and $\cos(\theta)$ are both rational. It is also very interesting that over finite fields, the problem of solving the equation of an elliptic curve becomes complicated, and we don't even have access to any analytic functions we can use to help us parametrize the solutions. Very shockingly, the fact that topologically over $\mathbb{C}$ the solutions look like a torus actually have some very serious consequences on the solutions of the elliptic curve over finite fields, but this is very much beyond these notes. Another surprising fact is that we have an algebraic group structure on our elliptic curve which is the popular construction made from intersecting lines with the elliptic curve. It turns out that $\wp$ not only forms a parametrization of the elliptic curve, but in fact, also forms a group isomorphism of the elliptic curve group law with the natural addition law of $\mathbb{C}/L$! The second group is taken by adding two points in the fundamental parallelogram, adding them usually, then translating to put it back in the fundamental parallelogram if necessary. One nice application is to find all n -torsion points of an elliptic curve: we just find the corresponding lattice and take the parametrization of the values $z = (a\omega_1 + b\omega_2)/n$ for $a, b \in \mathbb{Z}$ and ω_1, ω_2 a pair of generators for the fundamental parallelogram. Hence, there are n^2 n -torsion points, and in fact, the n -torsion points are isomorphic as a group to $(\mathbb{Z}/n\mathbb{Z})^2$. As an example of relating this to finite fields, it can be shown that over the algebraic closure of the finite field of characteristic p , the n -torsion points are also isomorphic to $(\mathbb{Z}/n\mathbb{Z})^2$.